Data Architecture in Data Engineering

Group 4

GUI KAH SIN (A23CS0080)
SABRINA HENG WEI QI (A23CS0265)

What is Data Architecture ?

DA = How data flow in a system

Collection

Storage

Tranformation

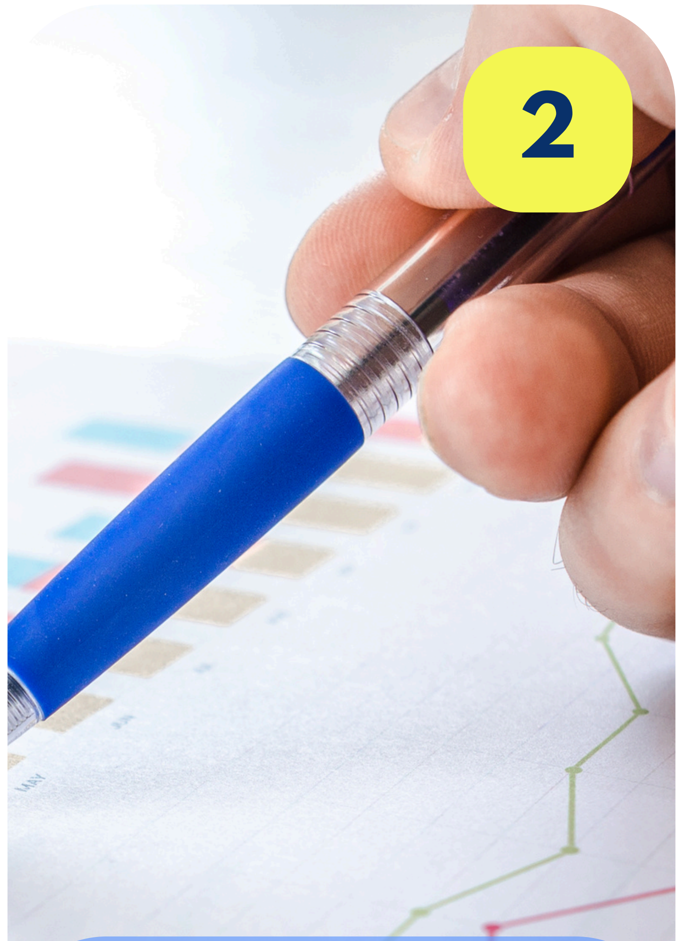
End Users

Background

Data Storage

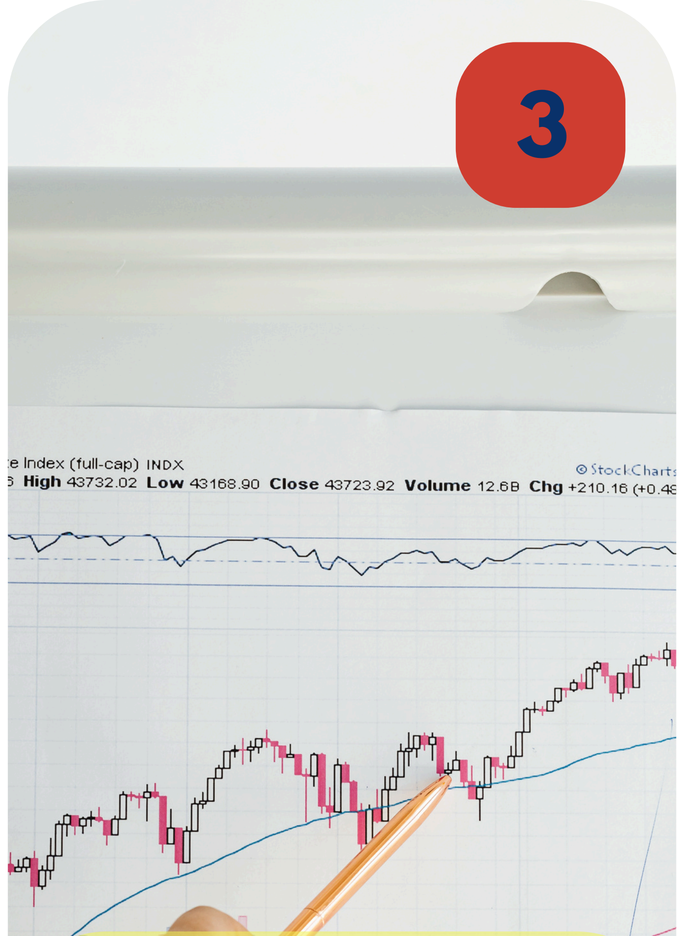
Data Warehouse

- store structured data
- not flexible



Data Lake

- store large amout raw data
- messy



Data Lakehouse

- combine both
- struture and flexible

Analysis & Discussion

How it works ?

Data flows:

Source → Ingestion → Storage → Processing → Serving

Applications

- Netflix → Recommendation system
- Amazon → Customer analytics

Challenges & Trends

- Data silos, security
- Data lakehouse, date mesh., cloud

Advantages

- Scalable & flexible
- Better decision-making

Disadvantages

- Complex design
- High cost

Recommendations

There are five suggestions that organizations can apply

Use Layered Architecture

It organizes system into parts

Adopt Cloud Platforms

Cloud gives flexible scalble resources

Implement Data Governance

Setting up rules and control to ensure data is accurate, secure, and well-managed

Use modern Processing Tools

Powerful tools process data faster

Train Skilled Data Engineers

Technologiy is constantly evolving

Conclusion

Data architecture is essential in data engineering as it enables **scalable**, **efficient**, and **data-driven decision-making** in modern organizations.